

llmXive Automated Review of DataComp-VLM: Improved Open Datasets for Vision-Language Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a comprehensive study on data mixing for VLMs, but several factual claims require verification against the provided evidence to ensure accuracy.

First, the abstract states the corpus contains “6T multimodal tokens,” and Section 4.1 confirms “6.0T tokens.” However, the breakdown of the 160 datasets (13 image-caption, 5 multimodal, 33 text-only, 109 instruction) and the sample count (3.9B) suggests a complex aggregation. While the numbers are internally consistent in the text, the abstract’s singular “6T” figure could be misinterpreted as a simple sum without the context of the specific tokenization and dataset inclusion criteria detailed in Section 4.1.

Second, the language composition in Section 4.1 cites “91.1% (Lingua) / 92.8% (NLLB) English.” Presenting two distinct percentages for the same attribute without clarifying the difference in measurement methodology or dataset subsets creates ambiguity. The reader cannot determine which figure represents the actual training distribution used for the reported results.

Third, the headline result in the abstract and Table 1 is an “8B VLM” achieving “63.6% accuracy” on a “33-task core suite.” However, the per-dataset breakdown tables (e.g., Table 6 for General Understanding) show individual scores like 73.0 for “Gen” and 73.0 for “Know.” The “Core Avg” of 63.6 is significantly lower than these component averages, implying a specific weighting or inclusion of lower-performing categories (like OCR or Multilingual) that drags the average down. The abstract’s phrasing “63.6% accuracy” without specifying the suite composition or the fact that it is a weighted average of disparate categories risks overstating the model’s performance on general tasks.

Fourth, the decontamination section claims a global removal rate of “0.29% of all training samples.” Given that specific datasets like InfoVQA lose 100% and ScienceQA 66.4%, this low aggregate figure suggests the removed datasets are a tiny fraction of the total pool volume. While mathematically possible, the stark contrast between the high per-dataset removal rates and the low global rate warrants a brief clarification in the text to prevent the reader from assuming the decontamination was ineffective or that the high-removal datasets were negligible.

Finally, the bibliography and text reference models such as “GPT-5.5” and “Gemini-3.1-pro (2026).” As of the current date, these models do not exist. If these are used as baselines in the experiments, the paper relies on hallucinated or future-dated references, which invalidates the comparative claims. If they are hypothetical, they must be clearly marked as such. This is a critical issue for the reproducibility and validity of the benchmark comparisons.

Action items.

- **[writing]** The paper presents a comprehensive study on data mixing for VLMs, but several factual claims require verification against the provided evidence to ensure accuracy. First, the abstract states the corpus contains “6T multimodal tokens,” and Section 4.1 confirms “6.0T tokens.” However, the breakdown of the 160 datasets (13 image-caption, 5 multimodal, 33 text-only, 109 instruction) and the sample count (3.9B) suggests a complex aggregation. While the numbers are internally consistent in the text, t

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 is a clear and well-structured flowchart that effectively visualizes the data collection pipeline described in its caption. All steps, data types, and final corpus pools are legible, and the logical flow from sourcing to the final compute-scale pools is easy to follow.

Figure 2

The figure is visually clear and effectively communicates the dataset composition and performance gains. However, the caption contains significant text errors (missing model name ‘DCVLM’ and the word ‘four’) and fails to explicitly map the cited dataset references to the specific lines in the chart, reducing its standalone clarity.

Figure 3

The figure effectively visualizes the scaling grid and supports the caption’s claims about mixture performance. However, the legend is only present in the rightmost panel, and the abbreviations used in the legend (‘IC’, ‘IT’) do not match the notation in the caption (‘c’, ‘i’).

Figure 4

The figure is well-structured with clear axes and legends, but the caption contains a numerical inaccuracy: it claims a +2.0% gain over the baseline for the medium scale optimal configuration, whereas the plotted data shows a gain of approximately 1.5%.

Figure 5

The figure effectively visualizes the ablation study with clear color coding, but the caption contains a factual error regarding the count of positive-gain configurations (stating 3 when 4 are shown) and the y-axis labels are overly dense, reducing readability.

Figure 6

Figure 6 is a clear and well-constructed bar chart that effectively supports its caption’s claim. The axes, units, legend, and specific data values (including delta annotations) are all present, legible, and correctly labeled.

Figure 7

The figure clearly visualizes the performance difference between sampling strategies with helpful delta annotations, but the x-axis configuration labels are undefined and the specific gain values slightly exceed the range described in the caption.

Figure 8

The figure effectively communicates that $T=1$ is near-optimal, but the x-axis lacks explicit units for temperature, and the orange baseline line is visually ambiguous without further explanation in the caption or legend.

Figure 9

The figure effectively displays the performance comparison, but the title and caption contain significant mathematical ambiguities regarding ‘improvement’ (absolute points vs. relative percent) and missing subject names. Additionally, the title includes raw LaTeX artifacts.

Figure 10

The figure is a clear bar chart showing per-dataset score changes, but the caption contains a grammatical error (‘where appeared strongest’) and slightly misaligns with the specific dataset names shown (e.g., ‘ChartQA_TEST’ vs ‘ChartQA’).

Figure 11

The bar chart clearly displays the performance gains from upsampling OCR data, but the caption contains a missing model name at the end and the textual claim about ‘closing half the gap’ is mathematically ambiguous compared to the bar values.

Figure 12

Figure 12 effectively demonstrates that online filtering adds negligible overhead, with clear axes and data points. However, the x-axis label for the 3-filter configuration is slightly ambiguous compared to the precise wording in the caption.

Action items.

- **[writing]** Figure 2: The caption contains multiple instances of missing text where the model name ‘DCVLM-Baseline’ should appear (e.g., ‘-Baseline outperforms...’, ‘trained on -Baseline’, ‘beats an 8B model trained on FineVision... a 4 compute reduction’). The text ‘DCVLM-’ is missing before ‘Baseline’ and the word ‘four’ is missing before ‘compute reduction’.
- **[writing]** Figure 2: The caption cites specific dataset references (deshmukh2025nvidia, an2025llava, wiedm...))
- **[writing]** Figure 2: The caption cites specific dataset references (deshmukh2025nvidia, an2025llava, wiedmann2025finevision) that are not defined or explained in the caption itself,

making it unclear which lines in the chart correspond to which dataset.

- **[science]** Figure 2: The x-axis label ‘Params, Tokens’ is ambiguous; it is unclear if the values represent parameter count, token count, or a combination, and the specific mapping of each point (e.g., ‘1B, 6.25B’) to these metrics is not explicitly defined in the axis or caption.
- **[writing]** Figure 3: The legend in the 4B Model panel uses ‘IC’ and ‘IT’ abbreviations (e.g., ‘10% IC, 70% IT’), but the caption defines the mixtures using ‘c’ and ‘i’ (e.g., ‘10c-70i’). The legend should be updated to match the caption’s terminology for consistency.
- **[writing]** Figure 3: The legend in the 4B Model panel is not visible in the other two panels (1B and 2B), forcing the reader to look at the far right panel to identify the lines in the left panels. A legend should be present in each subplot or the layout should be adjusted.
- **[science]** Figure 4: The caption claims the orange ring highlights the best configuration at each scale, but the left plot’s ring highlights 15% IT (43.7%) while the data shows 40% IT (43.4%) is lower; however, the ring is at 15% IT which is indeed the highest point shown. Wait, looking closer at the left plot, the point at 15% IT is ~43.7%, and the point at 40% IT is ~43.4%. The ring is correctly placed on the highest point. However, the caption states ‘At small scale, the 65%IC baseline is competitive’.
- **[writing]** Figure 4: The x-axis label ‘Instruction-Tuning %’ is clear, but the caption refers to ‘IC% <-> IT%’, implying a trade-off. The axis only shows IT%, which is fine, but the caption’s mention of ‘65%IC baseline’ requires the reader to infer that 65% IC = 35% IT. This is not explicitly stated in the figure or caption, though it is a reasonable assumption. However, the caption says ‘9-point sweep’, and counting the points on the x-axis: 15, 20, 25, 30, 35, 40, 50, 60, 70 — that is 9 points. So that i
- **[science]** Figure 4: The caption states ‘+2.0% over baseline’ for the medium scale optimal (70% IT), but visually comparing the 70% IT point (56.3%) to the 35% IT point (baseline, ~54.8%) yields a difference of approximately 1.5%, not 2.0%. This numerical discrepancy between the caption and the plotted data undermines the claim.
- **[writing]** Figure 5: The caption states ‘Only 3 of 39 configurations show gains,’ but the chart displays four bars with positive values (Mixtral >0.9, Nemotron >0.9, Nemotron >0.9, Core-benchmaxxed), creating a contradiction between the text and the visual data.
- **[writing]** Figure 5: The caption claims gains are ‘all from text quality classifiers applied exclusively to the text-only data pool,’ but the chart shows ‘Mixtral >0.9 (text only)’ and ‘Nemotron >0.9 (text only)’ as blue bars (Text-only modality), while ‘Core-benchmaxxed (text only)’ is also blue; however, the caption’s phrasing implies a specific subset that might be confusingly mapped to the ‘Text-only (T)’ legend entry which includes other negative bars.
- **[science]** Figure 5: The y-axis labels are extremely dense and overlapping (e.g., ‘Length text only (rm short <0.1)’), making specific filter configurations difficult to read and distinguish without zooming.

- **[writing]** Figure 7: The x-axis labels (e.g., ‘65c no-filter’, ‘65c + Mixtral’) are not defined in the caption or figure; the meaning of ‘65c’, ‘40c’, and ‘+ Mixtral’ is unclear without external context.
- **[science]** Figure 7: The green delta values (+2.2, +1.1, etc.) indicate the improvement of Sample-Level over Shard-Level, but the y-axis shows absolute ‘Micro Avg (%)’ values. The caption claims a ‘+1–2%’ improvement, yet the first bar shows a +2.2% gain, which slightly exceeds the stated range.
- **[writing]** Figure 8: The x-axis label ‘Temperature (T)’ is ambiguous because the tick labels (0.5, 0.8, 1.0, 2.0, 4.0) do not explicitly state the unit or scale (e.g., ‘ $\times 10^3$ ’ or ‘seconds’), and the caption does not define the temperature scale.
- **[writing]** Figure 8: The orange line segment labeled ‘T=1 baseline’ extends beyond the data point at T=1.0 without a clear endpoint or legend entry explaining its purpose, which may confuse readers about whether it represents a trend or a reference line.
- **[science]** Figure 9: The title claims a ‘+5.8% Micro Avg improvement’, but the chart shows ‘Ours’ (58.9) is only +5.8 points higher than ‘FineVision’ (53.1). If these are percentages, the relative improvement is ~10.9%, not 5.8%. If 5.8 is the absolute point difference, labeling it as a ‘%’ improvement in the title is misleading.
- **[writing]** Figure 9: The title contains raw LaTeX formatting artifacts (‘4B x 100B’ rendered as ‘4B x 100B’ with potential encoding issues or just poor typography) and the caption text ‘outperforms on 5 of 6 categories’ is missing the subject name (likely ‘DCVLM’ or ‘Ours’) due to a copy-paste error in the manuscript.
- **[science]** Figure 9: The caption claims a +13.2% gain in ‘General’ and +18.1% in ‘Vision’, but the chart shows absolute point gains of 13.2 (68.4-55.2) and 18.1 (57.2-39.1). Describing absolute percentage point differences as ‘% improvement’ is ambiguous and often mathematically incorrect (e.g., 13.2 points on a 55.2 base is a 24% relative increase).
- **[writing]** Figure 10: The caption states ‘The largest drops are on ScienceQA (-10.7%), nq (-7.6%), and ChartQA (-5.9%)’, but the chart shows ‘nq’ at -7.6% and ‘ChartQA_TEST’ at -5.9%, while ‘AI2D_TEST_NO_MASK’ is at -3.9%. The caption omits ‘AI2D_TEST_NO_MASK’ which is visually distinct, and the text ‘appeared strongest’ is grammatically incomplete (missing subject, likely ‘DataComp-VLM’).
- **[writing]** Figure 10: The caption text ‘categories where appeared strongest’ is missing the subject (e.g., ‘where DataComp-VLM appeared strongest’), making the sentence grammatically incomplete and confusing.
- **[writing]** Figure 11: The caption contains a sentence fragment at the end (‘closing roughly half the remaining gap with decontaminated.’) where the model name (likely ‘Baseline’) is missing.
- **[science]** Figure 11: The title claims a ‘+3.1% gain’ from upsampling, but the visual difference between the ‘Ours (standard)’ bar (45.8) and the ‘+ OCR Upsampled 25%’ bar (48.9) is 3.1

points; however, the caption implies this closes the gap with ‘decontaminated’ (51.7), yet the gap between 48.9 and 51.7 is 2.8, which is not ‘roughly half’ of the total gap (51.7 - 45.8 = 5.9). The math in the caption is slightly confusing or imprecise relative to the bars shown.

- **[writing]** Figure 12: The rightmost bar is labeled ‘3 filters (80%)’, but the caption states ‘Scaling to 3 filters at 80% rejection each’. The x-axis label is ambiguous and could be misinterpreted as a single filter with 80% rejection; it should explicitly state ‘3 filters (80% each)’ to match the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-written for its target audience, but it relies on several acronyms and subfield-specific terms that are not expanded at their first occurrence, potentially creating friction for a competent reader from an adjacent field (e.g., a computer vision researcher less familiar with NLP data curation, or vice versa).

Specifically, the term “pp” (percentage points) in the abstract and results sections is used without definition. While intuitive to many, it is a specific statistical shorthand that should be explicitly defined (e.g., “percentage points (pp)”) to avoid ambiguity with “parts per” or other metrics.

The language detection metrics in Section 3.1 reference “Lingua” and “NLLB” without expansion. “NLLB” is a specific Meta project (No Language Left Behind), and “Lingua” is ambiguous without context (likely a specific library or model). These should be expanded to ensure the reader understands the source of the language statistics.

In the decontamination sections (4.1 and 4.2), the paper introduces “SSCD” (Self-Supervised Copy Detection) and “MinHash” without defining them. While FAISS is widely known in retrieval, SSCD is a more niche descriptor, and MinHash, while standard in NLP/IR, is not universal to all ML subfields. A brief gloss or expansion at first use would significantly improve accessibility for a cross-disciplinary reader.

Finally, ensure that “SFT” (Supervised Fine-Tuning) is expanded at its very first occurrence in the document. If it first appears in Section 5, it should be defined there, as it is a core methodological term.

These are minor, low-effort edits (adding parenthetical expansions) that would make the paper fully self-contained for the intended “adjacent-field PhD” audience.

Action items.

- **[writing]** The paper is generally well-written for its target audience, but it relies on several acronyms and subfield-specific terms that are not expanded at their first occurrence, potentially creating friction for a competent reader from an adjacent field (e.g., a computer vision

researcher less familiar with NLP data curation, or vice versa). Specifically, the term “pp” (percentage points) in the abstract and results sections is used without definition. While intuitive to many, it is a specific statistic

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is logically sound and internally consistent. The central thesis—that data mixing strategies (specifically instruction-heavy mixtures) are more impactful than quality filtering at scale—is supported by a coherent chain of evidence. The experimental design (scaling grids, ablation studies, and control experiments) directly addresses the premises required to draw the stated conclusions.

Specifically, the transition from the observation that filtering yields negligible gains (Section 4) to the conclusion that mixing is the primary lever for improvement (Abstract, Section 5) is valid because the authors explicitly control for compute and token budgets, isolating the variable of data composition. The control experiments in Section 6 (SFT transfer and backbone robustness) successfully close potential logical gaps regarding the generality of the findings, ensuring the conclusion isn’t an artifact of a specific training stage or model initialization.

Definitions of the evaluation suites (Extended, Core, Validation) are consistent throughout the text and tables. The decontamination protocol (Section 3) is rigorously defined and applied consistently to both image and text modalities, with the chosen thresholds (0.75 for SSCD, 0.55 for MinHash) justified by the provided analysis of false positives/negatives, preventing any logical leap regarding data contamination. There are no contradictions between the abstract’s claims and the body’s results, nor are there any instances where a conclusion requires an unstated assumption to hold. The numerical claims in the tables align with the textual descriptions of the results.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper generally maintains a strong alignment between its claims and the evidence provided, particularly in its detailed ablation studies and decontamination protocols. However, there are three instances where the framing in the abstract and conclusion slightly exceeds the specific scope of the experimental evidence.

First, the claim that “instruction-heavy mixtures scale better than caption-heavy ones” is presented as a general principle in the abstract. The evidence (Section 5, Figure 2) supports this robustly for the Qwen2.5-Base and Qwen2.5-Instruct backbones, but the paper does not test other model families (e.g., Llama-based or proprietary architectures). While the control experiments suggest robustness to the *type* of instruction initialization, the claim of general “scaling better” across all VLMs is not fully licensed by the single-family study. Narrowing this to “within the Qwen2.5 family” or adding a qualifier about the tested architectures would align the claim with the evidence.

Second, the abstract highlights the 63.6% accuracy on the 33-task suite as a result of the dataset (“The resulting DataComp-VLM-Baseline enables...”). While the dataset is the primary variable, the specific performance is inextricably linked to the specific training recipe (InternVL architecture, specific hyperparameters, image resolution) detailed in Appendix A. The phrasing risks implying that the dataset alone is sufficient to achieve this result on any 8B model, whereas the evidence only supports this for the specific InternVL training pipeline. Clarifying that this performance is achieved “using the InternVL training recipe” would be more precise.

Finally, the paper makes a deliberate methodological choice to exclude grounding benchmarks (RefCOCO, etc.) due to contamination risks (Section 2.1). While this is a sound scientific decision, the conclusion presents the results on the remaining 33 tasks as a broad validation of the dataset’s improvements. This creates a scope gap: the paper claims to benchmark “improved open datasets” but omits a significant domain (spatial grounding) where the data mixing strategy might behave differently. The paper should include a limitations statement explicitly acknowledging that the reported gains and the “instruction-heavy” finding have not been validated on grounding tasks, preventing readers from over-generalizing the dataset’s utility to all VLM capabilities.

These are primarily framing issues that can be resolved by tightening the language in the abstract and adding a specific limitations paragraph, rather than requiring new experiments.

Action items.

- **[writing]** Abstract claims ‘instruction-heavy mixtures scale better’ generally, but evidence (Sec 5) only tests Qwen2.5 backbones. Scope claim to ‘within Qwen2.5 family’ or add evidence from other architectures.
- **[writing]** Abstract states 63.6% accuracy is enabled by the dataset, but results (Table 1) depend on the specific InternVL training recipe. Clarify that this performance is specific to the InternVL architecture and recipe used.
- **[writing]** Paper excludes grounding benchmarks (Sec 2.1) yet presents results as comprehensive. Add a limitations paragraph stating that gains are unverified for spatial grounding tasks, preventing over-generalization.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a dataset curation and benchmarking study (DataComp-VLM) focused on improving Vision-Language Models through data mixing strategies. The work aggregates 160 existing public datasets (e.g., from Hugging Face, academic repositories) and performs extensive decontamination and evaluation.

From a safety and ethics perspective, the paper is low-risk. The primary data sources are publicly available, and the authors explicitly address data provenance and licensing in the appendices (e.g., Section e003), noting specific licenses (CC-BY, Apache-2.0, etc.) and potential restrictions for certain datasets (e.g., LLaVA-Med, MultiUI). The paper does not involve collecting new human-subject data, nor does it release any Personally Identifiable Information (PII); the decontamination process described (Section e001) is designed to remove overlaps with evaluation sets, further reducing privacy risks.

The methodology does not introduce dual-use capabilities that lower the barrier to harm (e.g., automated vulnerability discovery or persuasive disinformation generation). The “safety” benchmarks mentioned are used for evaluation, not for generating harmful content. The paper appropriately excludes grounding benchmarks (RefCOCO) due to contamination risks, demonstrating a rigorous approach to validity rather than a safety failure.

No specific, foreseeable, non-trivial risks of harm were identified that are unaddressed in the text. The standard disclosures regarding data licensing and the use of public datasets appear sufficient for this type of research.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling benchmark and a strong central finding regarding data mixing, but the evidentiary design for the headline quantitative claims requires strengthening to rule out noise and confounds.

First, the primary claim of a +5.4pp improvement over FineVision (Abstract, Table 1) rests on single-point estimates without any measure of variance. In large-scale VLM training, performance can fluctuate significantly across random seeds (often >1-2pp). Without reporting results across at least 3-5 seeds with standard deviations or confidence intervals, it is impossible to determine if this gap represents a robust effect of the proposed data mixture or simply a lucky initialization. The current presentation treats a single run as a generalizable fact, which is insufficient for a benchmark paper.

Second, the conclusion that “filtering rarely helps” (Section 4.2) is potentially confounded by the experimental design. The filtering experiments remove data, thereby changing the total token count and the specific distribution of remaining samples compared to the “No Filter” baseline. A

performance drop (or lack of gain) could be due to the reduction in data volume or the specific removal of certain samples, rather than the act of filtering itself. To isolate the mechanism, the authors should include a control experiment where the unfiltered pool is subsampled to match the exact token count and distribution of the filtered pool, ensuring the comparison is strictly about the *quality* of the selected data, not the quantity.

Finally, the attribution of gains to “instruction-heavy mixtures” (Section 5) is complicated by potential compute mismatches. If the baseline models (e.g., FineVision) were trained on different token budgets or with different scaling laws than the DCVLM models, the observed improvement could be driven by the increased scale of the DCVLM training rather than the mixture strategy. The paper needs to explicitly demonstrate that the gains hold when comparing mixtures at identical compute budgets (token counts) to rule out scale as the primary driver.

Addressing these design gaps—specifically by adding seed variance, a compute-matched subsampling control, and explicit token-budget ablations—would solidify the causal claims regarding data mixing and filtering.

Action items.

- **[writing]** The paper presents a compelling benchmark and a strong central finding regarding data mixing, but the evidentiary design for the headline quantitative claims requires strengthening to rule out noise and confounds. First, the primary claim of a +5.4pp improvement over FineVision (Abstract, Table 1) rests on single-point estimates without any measure of variance. In large-scale VLM training, performance can fluctuate significantly across random seeds (often >1-2pp). Without reporting results across

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical reporting in this paper is generally consistent with current ML conventions regarding the presentation of scaling laws and benchmark results, but it lacks necessary uncertainty quantification for its primary quantitative claims.

The most significant omission is the absence of variance reporting for the main results. In Section 4.2 and Table 1, the authors report specific accuracy percentages (e.g., “63.6% accuracy”) for the DCVLM-Baseline and comparisons against FineVision. These are presented as single point estimates. While the text acknowledges that POPE was removed from the suite due to high seed variance (“16% seed variance”), it does not provide the standard deviation, standard error, or confidence intervals for the remaining 33 tasks or the aggregate scores. In deep learning experiments, especially with large models and limited compute budgets, run-to-run variance can be substantial. Reporting a single number without a spread (e.g., mean $\pm$ SD over 3-5 seeds) makes it impossible for the reader to assess whether the reported +5.4pp improvement over FineVision is a robust effect or potentially within the noise of the training process. The authors

should aggregate results from multiple seeds (if available) or explicitly state the number of seeds used and report the variance.

Additionally, the selection of the optimal learning rate in Section 4.1 (Table 2) is based on identifying the maximum mean score across 5 values. No statistical test is performed to determine if the differences between the top-performing LR and the others are statistically significant. Given the small sample size (likely 1-3 runs per cell), the “optimal” LR might be a result of random fluctuation. While this is a common practice in the field, the paper would be more rigorous if it reported the standard deviation for each LR setting or acknowledged the lack of statistical significance testing for the hyperparameter sweep.

Finally, the claim of “near-perfect fidelity” ($r=0.99$) in the control experiments (Section 5.1) is supported by a high correlation coefficient, but the paper does not report the confidence interval for this correlation or the p-value. While $r=0.99$ with $N=27$ is clearly significant, providing the 95% CI would strengthen the claim of “near-perfect” by quantifying the precision of the estimate.

These are primarily reporting issues (writing severity) rather than fundamental flaws in the experimental design or analysis logic, as the field often accepts mean-over-seeds without formal hypothesis tests. However, the complete absence of any uncertainty metric for the headline results is a gap that should be filled to ensure the numbers mean what the paper claims they mean.

Action items.

- **[writing]** Section 4.2 and Table 1 report single-point accuracy scores (e.g., 63.6%) for the proposed method and baselines without any measure of uncertainty (SD, SE, or CI). While the text mentions removing POPE due to ‘16% seed variance,’ it fails to report the variance for the main results. Report mean $\pm$ SD over at least 3 seeds for all key comparisons to distinguish signal from noise.
- **[writing]** Section 4.1 (Learning Rate Sweep) and Table 2 present results for 5 LR values across 3 model scales. The optimal LR is selected based on the highest mean score without any statistical test (e.g., ANOVA or post-hoc t-tests) to confirm the difference is significant. Given the small number of runs implied, the ‘optimal’ choice may be noise. Report the standard deviation across seeds for each LR or apply a correction for multiple comparisons if claiming one LR is definitively better.
- **[writing]** Section 5.1 claims ‘near-perfect fidelity’ (Pearson $r=0.99$) between pretraining and post-SFT performance based on 27 checkpoints. While the correlation is high, the paper does not report the confidence interval for the correlation coefficient or the p-value. With $N=27$, a correlation of 0.99 is significant, but the precision of this estimate should be quantified to support the ‘near-perfect’ claim.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and the prose is clear, but there are several instances where the flow is interrupted by abrupt transitions or missing signposting, particularly in the introductory and methodological sections.

The **Contributions** section (Section 1) and **Extended Related Work** (Section 2) suffer from a lack of narrative framing. Both sections open immediately with lists or subsection headers (`\smallsec`) without a topic sentence to guide the reader. In Section 1, the reader is presented with a list of names and roles without a sentence explaining that this section details the division of labor. Similarly, Section 2 dives into specific sub-topics without a brief overview of how these topics relate to the paper’s specific contributions. Adding a single introductory sentence to each section would significantly improve the “hand-off” to the detailed content.

In **Section 3 (Train-Test Decontamination)**, the transition between the description of the matching procedure and the justification for the threshold choice is slightly abrupt. The subsection “Choosing the threshold” starts with “We use a similarity threshold of 0.75...” which feels like a sudden declaration. A smoother transition would explicitly link the previous paragraph’s description of the method to the decision-making process for the parameters (e.g., “Having established the matching procedure, we now determine the optimal threshold...”).

The **Abstract** is concise but could be slightly more self-contained. While it mentions the improvement over FineVision, providing the absolute baseline score in the abstract would allow a reader to immediately grasp the magnitude of the result without needing to scan the body text for the comparison point.

Overall, the writing is competent, but these minor structural adjustments would ensure the reader never has to pause to infer the purpose of a section or the logical connection between paragraphs.

Action items.

- **[writing]** The paper is generally well-structured and the prose is clear, but there are several instances where the flow is interrupted by abrupt transitions or missing signposting, particularly in the introductory and methodological sections. The Contributions section (Section 1) and Extended Related Work (Section 2) suffer from a lack of narrative framing. Both sections open immediately with lists or subsection headers (`\smallsec`) without a topic sentence to guide the reader. In Section 1, the reader is